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Connecting Mechanism and an Umbrella Thereof

Abstract

A connecting mechanism for connecting a long rod and a short rod of an umbrella comprises a rod connector and two detachable decorating parts. The rod connector has a long groove at a top of the rod connector for receiving the long rod. The rod connector is connected to a middle portion of the long rod. The rod connector has a U-shaped groove at a bottom of the rod connector for receiving an end of the short rod. The end of the short rod is rotatably connected to the rod connector through a connecting pin. The two decorating parts are respectively disposed at a front side and a back side of the rod connector to shelter the connecting pieces such as bolts, so that the overall appearance is improved. Moreover, the rod connector is easy to mount, and the decorating parts are detachable and easy to replace.

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Background/Summary

TECHNICAL FIELD OF THE INVENTION

[0001] The present invention relates to a field of outdoor umbrellas, and relates to an umbrella, in particular to a connecting mechanism for connecting a long rod and a short rod of an umbrella and an umbrella thereof.

BACKGROUND OF THE INVENTION

[0002] As a kind of outdoor leisure items, umbrellas are widely applied in squares, beaches, parks, gardens or similar places to shield an individual from sun, wind and rain. The existing umbrellas in the market generally comprise a base, a column and an umbrella frame, wherein the umbrella frame comprises an upper nest, a lower nest, a plurality of long rods and short rods, and each long rod and the corresponding short rod are usually connected by a rod connector. However, the rod connector has bolt holes on a surface of the rod connector, so it is not aesthetic.

[0003] Upon examination, a Chinese Patent CN203182190U (patent No.: CN201320188172.7) discloses a rod connecting device, comprising a first clamping arm, a second clamping arm, a connecting portion and a fastener, wherein the first clamping arm and the second clamping arm are oppositely arranged; the connecting portion is disposed between the first clamping arm and the second clamping arm, and connected to the first clamping arm and the second clamping arm, respectively, so as to divide the space between the first clamping arm and the second clamping arm into two parts; the fastener runs through the connecting portion; a first rod pin end accommodating portion is disposed on an inner wall of the first clamping arm, a second rod pin end accommodating portion is disposed on an inner wall of the second clamping arm, and the first rod pin end accommodating portion and the second rod pin end accommodating portion are oppositely arranged; the first clamping arm has an inclined surface on the inner wall of the first clamping arm; and, the inclined surface extends from an outer periphery of the inner wall of the first clamping arm to the first rod pin end accommodating portion and is inclined toward the inner wall of the second clamping arm. This rod connecting device is simple, but the short rods cannot be mounted stably and firmly, so that the connection is easy to fail after long-time use and wear.

SUMMARY OF THE INVENTION

[0004] It is a first object of the present invention to provide a connecting mechanism for connecting a long rod and a short rod of an umbrella, which has a simple and reasonable design of structure, a firm and stable connection and an aesthetic appearance.

[0005] It is a second object of the present invention to provide an umbrella with simple and reasonable design of structure, easy operation and aesthetic appearance.

[0006] For achieving the first object, the connecting mechanism for connecting a long rod and a short rod of an umbrella comprises a rod connector having a top, a bottom, a front side and a back side; two detachable decorating parts; wherein, the rod connector has a long groove at the top of the rod connector for receiving the long rod, the rod connector is connected to a middle portion of the long rod; the rod connector has a U-shaped groove at the bottom of the rod connector for receiving an end of the short rod, the end of the short rod is rotatably connected to the rod connector through a connecting pin; the two detachable decorating parts are respectively disposed at the front side and the back side of the rod connector.

[0007] As an improvement, the rod connector comprises two side plates vertically disposed and a transverse plate transversely disposed between the two side plates which connects middle portions of the two side plates; the long groove is defined above the transverse plate and between the two side plates, the U-shaped groove is defined beneath the transverse plate and between the two side plates.

[0008] Preferably, the rod connector is manufactured into an integral part by injection molding, die

casting or stamping, which is easy for production and assembly.

[0009] Preferably, each side plate has a protruding edge and a recess enclosed by the protruding edge at an outer surface of each side plate; each decorating part has a periphery which is fitted to the corresponding side plate, each decorating part has a convex surface protruding into the recess of the corresponding side plate, each decorating part has one or more elastic buckles protruding from the convex surface, and each side plate has one or more corresponding positioning holes at the recess each for receiving one elastic buckle of each decorating part; each decorating part is positioned on one corresponding side plate by pressing the one or more elastic buckles into the corresponding positioning holes of each side plate.

[0010] Preferably, each side plate has one or more connecting counterbores at an upper portion of the outer surface of each side plate for receiving one or more connecting pieces, and has a pin hole at a lower portion of the outer surface of each side plate for receiving the connecting pin; each decorating part has a plurality of evading notches on the convex surface corresponding to the one or more connecting counterbores and the pin hole.

[0011] Preferably, the number of the positioning holes is three and the three positioning holes are distributed symmetrically, one of the positioning holes is located in a middle of the upper portion of the outer surface of each side plate, and the other two positioning holes are located on the left and right sides of the lower portion of the outer surface of each side plate, and correspondingly, the number of the elastic buckles on the decorating parts is three.

[0012] Finally, an outer surface of each decorating part has leather texture, or the outer surface of each decorating part is covered with leather.

[0013] For achieving the second object, the umbrella comprises a column; an umbrella frame having a plurality of long rods, a plurality of short rods, an upper nest and a lower nest; wherein, each long rod is connected to the corresponding short rod through the above-stated connecting mechanism.

[0014] Preferably, the umbrella frame is a frame of a cantilever umbrella, detachable decorating parts are disposed on surfaces of hinges, inclined connecting blocks and/or sliders of the cantilever umbrella, the hinges, inclined connecting blocks and/or sliders have a plurality of positioning holes at peripheral edges of hardware fittings exposed on the surfaces of the hinges, inclined connecting blocks and/or sliders of the cantilever umbrella, a plurality of corresponding elastic buckles matched with the positioning holes are protruded from surfaces of the decorating parts, the decorating parts are disposed and positioned on the surfaces of the hinges, inclined connecting blocks and/or sliders of the cantilever umbrella by pressing the plurality of elastic buckles into the positioning holes.

[0015] Finally, the decorating parts on the surfaces of the hinges, inclined connecting blocks and/or sliders of the cantilever umbrella are of a circular, oval or strip shape, respectively, but are not limited to the three shapes.

[0016] Compared with the prior art, the connecting mechanism for connecting a long rod and a short rod of an umbrella and the umbrella thereof of the present invention has the following advantages. The two decorating parts are respectively disposed at the front side and the back side of the rod connector to shelter the connecting pieces such as bolts, so that the overall appearance is improved. Moreover, the rod connector is easy to mount, and the decorating parts are detachable and easy to replace. The connecting mechanism for connecting a long rod and a short rod of an umbrella of the present invention is simple and reasonable in structure and is firmly and stably connected to the rod, and the decorating parts are easy to disassemble and assemble, so that the overall appearance is greatly improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. **1** is a perspective view of an umbrella and an enlarged view of an umbrella with a connecting mechanism for connecting a long rod and a short rod according to Embodiment **1** of the present invention;

[0018] FIG. **2** is a perspective view of a decorating part according to Embodiment **1** of the present invention;

[0019] FIG. **3** is an exploded view of a rod connector and the decorating part according to Embodiment **1** of the present invention;

[0020] FIG. **4** is a front view of two different kinds of decorating parts according to Embodiment **1** of the present invention;

[0021] FIG. **5** is a sectional view of a second kind of the decorating part which is covered with leather;

[0022] FIG. **6A-6C** are respectively a perspective view of a hinge, an inclined connecting block and a slider of a cantilever umbrella before a decorating part is mounted on according to Embodiment **2** of the present invention;

[0023] FIG. **7A-7C** are a perspective view of three kinds of decorating parts with different shapes according to Embodiment **2** of the present invention;

[0024] FIG. **8A-8C** are respectively a perspective view of the hinge, inclined connecting block and slider of the cantilever umbrella after an adapted decorating part showing in FIG. **7A-7C** is mounted on according to Embodiment **2** of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0025] The present invention will be further described below in detail by embodiments with reference to the accompanying drawings.

Embodiment 1

[0026] As shown in FIGS. **1-5**, an umbrella comprises a column **8** and an umbrella frame. The umbrella frame comprises a plurality of long rods **4**, a plurality of short rods **5**, an upper nest **6** and a lower nest **7**. Each long rod **4** is connected to the corresponding short rod **5** through a connecting mechanism A of the present invention.

[0027] The connecting mechanism A for connecting the long rod **4** and the short rod **5** comprises a rod connector **1** having a top, a bottom, a front side and a back side, and two detachable decorating parts **2**. The rod connector **1** has a long groove **13** at the top of the rod connector **1** for receiving the long rod **4**. The rod connector **1** is connected to a middle portion of the long rod **4**. The rod connector **1** has a U-shaped groove **14** at the bottom of the rod connector **1** for receiving an end of the short rod **5**. The end of the short rod **5** is rotatably connected to the rod connector **1** through a connecting pin. The two detachable decorating parts **2** are respectively disposed at the front side and the back side of the rod connector **1**. The rod connector **1** comprises two side plates **11** vertically disposed and a transverse plate **12** transversely disposed between the two side plates **11** which connects middle portions of the two side plates **11**. The long groove **13** is defined above the transverse plate **12** and between the two side plates **11**. The U-shaped groove **14** is defined beneath the transverse plate **12** and between the two side plates **11**. The rod connector **1** is manufactured into an integral part by injection molding, easy for production and assembly. Each side plate **11** has a protruding edge and a recess **15** enclosed by the protruding edge at an outer surface of each side plate **11**. Each decorating part **2** has a periphery which is fitted to the corresponding side plate **11**. Each decorating part **2** has a convex surface **21** protruding into the recess **15** of the corresponding side plate **11**. Each decorating part **2** has one or more elastic buckles **22** protruding from the convex surface **21**, and each side plate **11** has one or more corresponding positioning holes **16** at the recess **15** each for receiving one elastic buckle **22** of each decorating part **2**. Each decorating part **2** is positioned on one corresponding side plate **11** by pressing the one or more elastic buckles **22** into the corresponding positioning holes **16** of each side plate **11**.

[0028] Each side plate **11** has one or more connecting counterbores **17** at an upper portion of the outer surface of each side plate **11** for receiving one or more connecting pieces (such as bolts, rivets), and has a pin hole **18** at a lower portion of the outer surface of each side plate **11** for receiving the connecting pin. Each decorating part **2** has a plurality of evading notches **23** on the convex surface **21** corresponding to the one or more connecting counterbores **17** and the pin hole **18**. The number of the positioning holes **16** in this embodiment is three and the three positioning holes **16** are distributed symmetrically, one of the positioning holes **16** is located in a middle of the upper portion of the outer surface of each side plate **11**, and the other two positioning holes **16** are located on the left and right sides of the lower portion of the outer surface of each side plate **11**, and correspondingly, the number of the elastic buckles **22** on the decorating parts **2** is three.

[0029] There are two kinds of decorating parts **2**, as shown in FIGS. **4** and **5**. As to one kind of decorating parts, an outer surface of each decorating part **2** has leather texture **2a**, and as to the other kind of decorating parts, the outer surface of each decorating part **2** is covered with a layer of leather **3**.

[0030] In this embodiment, each side plate **11** of the rod connector **1** has a hexagon shape, and the upper and lower portions of each side plate **11** are asymmetric. Similarly, the recess **15** is also a hexagon with asymmetric upper and lower portions. Each decorating part **2** has a periphery which is fitted to the corresponding side plate **11**. The rod connector **1** can also be in other shapes.

[0031] During mounting, the rod connector **1** is connected to the middle portion of the long rod **4**, and the end of the short rod **5** is inserted into the U-shaped groove **14** of the rod connector **1** and then connected through the connecting pin. Finally, the two detachable decorating parts **2** are respectively disposed at the front side and the back side of the rod connector **1**. Thus, assembling of the rod connecting mechanism is completed.

[0032] In the present invention, since the decorating parts **2** are respectively disposed at the front side and the back side of the rod connector **1**, the connecting pieces such as bolts (not shown in figures) can be sheltered, so that the overall appearance is improved. Moreover, the decorating parts **2** are detachable and easy to replace.

Embodiment 2

[0033] In this embodiment, the umbrella is a cantilever umbrella which has a hinge **10**, an inclined connecting block **20** and a slider **30**. As shown in FIGS. **8A-8C**, three kinds of detachable decorating parts a, b, c (as shown in FIGS. **7A-7C**) are disposed on surfaces of the hinge **10**, the inclined connecting block **20** and the slider **30** (as shown in FIGS. **6A-6C**) of the cantilever umbrella. The hinge **10** has a plurality of positioning holes **100** at an outer surface of hardware fittings of the hinge **10**, the inclined connecting block **20** has a plurality of positioning hole **200** at an outer surface of hardware fittings of the inclined connecting block **20**, and the slider **30** has a plurality of positioning hole **300** at an outer surface of hardware fittings of the slider **30**. A plurality of elastic buckles a**1**, b**1**, c**1** matched with the positioning holes **100**, **200**, **300** of the connecting objects (the hinge **10**, the inclined connecting block **20** and the slider **30**) are correspondingly protruded from the surfaces of the decorating parts a, b, c facing to the connecting objects. Each decorating part a, b, c is respectively positioned on the surfaces of the hinge **10**, inclined connecting block **20** and slider **30** of the cantilever umbrella by pressing the plurality of elastic buckles a**1**, b**1**, c**1** into the positioning holes **100**, **200**, **300**. In this embodiment, the decorating part a on the hinge **10** is of a circular shape, the decorating part b on the inclined connecting block **20** is of an oval shape, and the decorating part c on the slider **30** is of a strip shape.

[0034] The decorating parts a, b, c are not limited to the three kinds of shape as shown in FIGS. **7A-7C**, and can also be in other shapes.

[0035] In this embodiment, since the decorating parts a, b, c are respectively on the surfaces of the hinge **10**, inclined connecting block **20** and slider **30** of the cantilever umbrella, the connecting pieces such as bolts (as shown in FIGS. **6A-6C**) can be sheltered, so that the overall appearance is improved. Moreover, the decorating parts a, b, c are detachable and easy to replace.

[0036] The protection scope of the present invention is not limited to each embodiment described in this description. Any changes and replacements made on the basis of the scope of the present invention patent and of the description shall be included in the scope of the present invention patent.

Claims

1. A connecting mechanism for connecting a long rod and a short rod of an umbrella, comprising: a rod connector having a top, a bottom, a front side and a back side; two detachable decorating parts; wherein, the rod connector has a long groove at the top of the rod connector for receiving the long rod, the rod connector is connected to a middle portion of the long rod; the rod connector has a U-shaped groove at the bottom of the rod connector for receiving an end of the short rod, the end of the short rod is rotatably connected to the rod connector through a connecting pin; the two detachable decorating parts are respectively disposed at the front side and the back side of the rod connector.
2. The connecting mechanism of claim 1, wherein the rod connector comprises two side plates vertically disposed and a transverse plate transversely disposed between the two side plates which connects middle portions of the two side plates; the long groove is defined above the transverse plate and between the two side plates, the U-shaped groove is defined beneath the transverse plate and between the two side plates.
3. The connecting mechanism of claim 1, wherein the rod connector is manufactured into an integral part by injection molding, die casting or stamping.
4. The connecting mechanism of claim 2, wherein each side plate has a protruding edge and a recess enclosed by the protruding edge at an outer surface of each side plate; each decorating part has a periphery which is fitted to the corresponding side plate, each decorating part has a convex surface protruding into the recess of the corresponding side plate, each decorating part has one or more elastic buckles protruding from the convex surface, and each side plate has one or more corresponding positioning holes at the recess each for receiving one elastic buckle of each decorating part; each decorating part is positioned on one corresponding side plate by pressing the one or more elastic buckles into the corresponding positioning holes of each side plate.
5. The connecting mechanism of claim 4, wherein each side plate has one or more connecting counterbores at an upper portion of the outer surface of each side plate for receiving one or more connecting pieces, and has a pin hole at a lower portion of the outer surface of each side plate for receiving the connecting pin; each decorating part has a plurality of evading notches on the convex surface corresponding to the one or more connecting counterbores and the pin hole.
6. The connecting mechanism of claim 4, wherein the number of the positioning holes is three and the three positioning holes are distributed symmetrically, one of the positioning holes is located in a middle of the upper portion of the outer surface of each side plate, and the other two positioning holes are located on the left and right sides of the lower portion of the outer surface of each side plate, and correspondingly, the number of the elastic buckles on the decorating parts is three.
7. The connecting mechanism of claim 1, wherein an outer surface of each decorating part has leather texture, or the outer surface of each decorating part is covered with leather.
8. An umbrella, comprising: a column; an umbrella frame having a plurality of long rods, a plurality of short rods, an upper nest and a lower nest; wherein, each long rod is connected to the corresponding short rod through the connecting mechanism of claim 1.
9. The umbrella of claim 8, wherein the umbrella frame is a frame of a cantilever umbrella, detachable decorating parts are disposed on surfaces of hinges, inclined connecting blocks and/or sliders of the cantilever umbrella, the hinges, inclined connecting blocks and/or sliders have a plurality of positioning holes at peripheral edges of hardware fittings exposed on the surfaces of the hinges, inclined connecting blocks and/or sliders of the cantilever umbrella, a plurality of

corresponding elastic buckles matched with the positioning holes are protruded from surfaces of the decorating parts, the decorating parts are disposed and positioned on the surfaces of the hinges, inclined connecting blocks and/or sliders of the cantilever umbrella by pressing the plurality of elastic buckles into the positioning holes.

10. The umbrella of claim 9, wherein the decorating parts on the surfaces of the hinges, inclined connecting blocks and/or sliders of the cantilever umbrella are of a circular, oval or strip shape, respectively.
